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A Novel Electrolyte Solvent for Rechargeable Lithium and Lithium-Ion Batteries

S. S. Zhang and C. A. Angell*

Department of Chemistry, Arizona State University, Tempe, Arizona 85287-1604, USA

ABSTRACT

We describe a new type of electrolyte solvent which provides remarkable improvement in properties of electrolyte systems containing it. These solvents contain acidic boron atoms, in most cases linked into heterocycles by reaction of boric acid, or oxide, with glycols. Nicknamed BEG solvents (for boric acid esters of glycol), these show great propensity for dissolving salts, stabilizing alkali metals against corrosion, and in some cases, stabilizing other solvents (particularly alkene carbonates), against anodic decomposition. In the most favorable case so far, the 1,3 propylene glycol boric acid ester (BEG-1) which contains two linked borate groups, the mixed solvent formed by mixing 1 part BEG-1 with two parts ethylene carbonate provides an electrochemical stability window in excess of 5.8 V (cf. 4.5 V for ethylene carbonate alone with the same salt), an ambient temperature conductivity of $1.7 \times 10^{-3} \text{ S cm}^{-1}$ with 1 M LiClO₄, and enduring stability against metallic Li, which remains shiny on prolonged (days) immersion at 100°C. We present data on various electrolyte solutions containing these components and then show their utility in devices by rubberizing them with polymers and constructing voltaic cells Li/LiMn₂O₄ and C/LiMn₂O₄, which show excellent charge/discharge characteristics and cyclability.

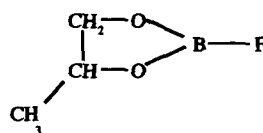
Introduction

Many nonaqueous electrolytes and polymer gel electrolytes, comprising polar organic solvents and their mixtures and having ambient ionic conductivities above 10^{-3} S/cm , have been developed for use with rechargeable lithium and carbonaceous anode lithium-ion batteries. Unfortunately, metallic lithium reacts with almost all organic solvents to form a low-conductivity passivation layer on the surface of the lithium, and repeated cycling frequently produces dendritic deposits which eventually short-circuit the batteries. Batteries using inorganic chloride electrolytes, such as Li/SOCl₂ and Li/SO₂Cl₂ batteries, suffer from voltage delay problems due to the formation of a polycrystalline LiCl passivation layer on the lithium surface. As a result of such electrolyte problems, the development of rechargeable lithium batteries has been greatly inhibited.

A wide variety of approaches has been used in attempts to improve lithium deposition morphology and to increase lithium cycling efficiency, with little success so far. Typical examples have been the use of organic or inorganic additives,¹⁻¹² introduction of new salts,¹³⁻¹⁷ Li alloys,¹⁸⁻²⁰ and finally, the lithium-free carbon anode.²¹⁻²⁴ Even in the case of carbonaceous anode lithium-ion batteries, in which the use of metal lithium is completely avoided, there remains an electrolyte problem limiting development. It is found that the common electrolyte solvents, such as propylene carbonate (PC) and ethylene carbonate (EC), easily cointercalate into carbon layers and decompose on the surface of fresh carbon particles (probably because of the electrocatalytic nature of fresh carbon structure). This leads to an

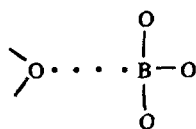
irreversibility of the first cycle, requiring extra lithium-rich cathode material (LiCoO₂, LiNiO₂, or LiMnO₂) to be included in the cells in order to compensate for the consumption in the initial irreversible cycle. To minimize this irreversibility of initial lithium intercalation, the use of co-solvents²⁵⁻²⁷ and the modification of carbonaceous structure²⁸⁻³¹ have been investigated. However, none of the above-mentioned methods is suitable for both lithium anode and carbon anode. Thus, a great need exists for new solvents and co-solvents with special stability in the presence of lithium metal. This in effect requires an ability of the solvent to react with metallic lithium in a benign way, i.e., so as to generate a stable Li⁺-conducting, nonpropagating film on the lithium surface. We report here on a class of molecular substance which appears to behave in this desirable manner and has remarkable stabilizing effects on co-solvents at positive potentials as well.

Our original motivation in this work was to find a molecular liquid which would dissolve salts by solvating the anions in order that the alkali metal transport number could become more favorable. To this end, we prepared initially the boron analog of propylene carbonate, viz.

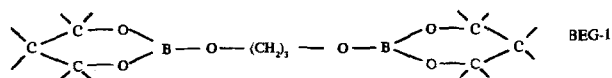


with the idea that the under-coordinated boron would seek a fourth ligand from the salt anion. Indeed, this liquid proves to be an excellent solvent for salts. However, it is a much more viscous liquid than PC, presumably due to intermolecular interactions of the type

* Electrochemical Society Active Member.



As a consequence, the solutions are relatively poor conductors. Seeking to enhance the anion solvation, we then examined the twin borate molecule described by Steinberg³²



We have prepared and tested several variants of these BEG solvents,^{33,34} but for the present article, we focus on the particular case designated BEG-1 and its molecular solutions with EC and/or PC. We describe its preparation, the testing for stability against metallic Li, and the properties of various salt solutions in it, both in BEG-1 alone and in mixed solvents BEG-1 plus alkene carbonates. Finally, we describe the preparation and performances of different rechargeable lithium cells based on these solutions.

Experimental

Materials.—Propylene carbonate (Aldrich, 99%) and ethylene carbonate (Aldrich, 98%) were heated over lithium at 100°C for 6 h and then distilled under the reduced pressure. Dimethyl ether (DME), SOCl_2 , SO_2Cl_2 , and POCl_3 (all from Aldrich) were refluxed over lithium for 5 h and distilled. $\text{CH}_3\text{SO}_3\text{Cl}$ (Aldrich, 99%) was directly distilled prior to use. Poly(ethylene oxide) molecular weight (Mw) 8×10^6 , poly(vinyl acetate) Mw 140,000, and poly(methylene methacrylate) Mw 996,000 were purchased from Aldrich and dried at 60°C under vacuum for 36 h. Lithium salts, LiClO_4 , $\text{CF}_3\text{SO}_3\text{Li}$ (Aldrich), and $\text{LiN}(\text{SO}_2\text{CF}_3)_2$ (donated by 3M) were dried at 150°C for 24 h under vacuum prior to use.

BEG-1 was prepared by reacting boric acid and 1,3-propanediol as described in the literature³² and in more detail in Ref. 33 and 34. BEG-1 is a colorless, viscous liquid with a boiling point of 125 to 128°C/0.05 mmHg and a glass transition temperature of ca. -80°C. It has a high capacity for solution of lithium salts. LiPF_6 is an exception; its low solubility is explained by the low electron donor capacity of the perfluorated anion. All the electrolyte solutions were prepared by warm dissolution of anhydrous salts in the BEG or BEG-1:EC mixtures in a nitrogen-filled dry box. The concentration unit used is the molality, m .

Polymer "gels," i.e., rubbery solid solutions, were made as follows: the appropriate amount [usually 10 to 20 weight percent (w/o)] of high molecular weight polymer [usually poly(methyl methacrylate) (PMMA) 10^6] was weighed into the electrolyte solution, and the mixed slurry left in a Pyrex bottle at 80 to 90°C for 10 h, during which the polymer dissolved completely to yield a transparent, colorless, viscous melt. The melt was sandwiched between TeflonTM-coated paper sheets and pressed with two steel plates. Upon cooling, a self-supporting polymer gel film was formed which could be stretched like rubber. The rubber-liquid transition temperature^{35,36} falls between ambient and 80°C. For cell tests, a solution which was 1.0 m LiClO_4 in the mixed solvent consisting of 1:2 (by weight) BEG-1:EC was used in either liquid or polymer gel electrolyte form.

LiMnO_2 for use as cathode material in the test cells was made by reacting $\text{LiOH} \cdot \text{H}_2\text{O}$ (Aldrich, 98%) and Mn_2O_3 (Aldrich, 99%) in a 2:1 molar ratio at 700°C for 24 h in air. The resultant LiMnO_2 is a brown-black powder.

Carbon containing 4 to 5 w/o nitrogen for use in cell anodes was prepared from aniline-aldehyde resin by pyrolysis at 900°C for 24 h in an inert gas. The pyrolyzed product was ground into a fine powder and mixed with 25 w/o polymer gel electrolyte.

Lithium metal-electrolyte solvent stability tests.—To explore the possibility of practical use of BEG-1 solvent in

lithium batteries, ~0.5 cm^2 pieces of freshly exposed, shiny Li foil were added to vials containing, respectively, pure PC, 1:9 BEG-1:PC, and 15:85 BEG-1:PC mixtures using inert atmosphere procedures. The vials were heated at 95°C for 24 h.

Conductivity measurements.—Ionic conductivities of the electrolyte solutions and gels were determined from complex impedance plots obtained using a Hewlett Packard 4192A LF impedance analyzer over the frequency range 5 Hz to 13 MHz. A twin platinum electrode dip-type cell with fixed cell constant of 2.0 cm^{-1} was used for each measurement. The rate of temperature rise was controlled at 1°C/min by means of a computer under control of house software.

Electrochemical stability.—A Princeton Applied Research Corporation potentiostat/galvanostat Model 273 was used to perform cyclic voltammetry experiments under software control. The scan rate was 10 mV/s. The cell comprised three platinum electrodes, two of which served as working and counterelectrodes, the third acting as a pseudoreference electrode. An additional experiment verified that except for a potential shift of ~3 V, there is no difference in either the cyclic voltogram shape or the electrochemical window of the electrolyte, obtained when lithium is substituted for platinum as the reference electrode. The use of platinum is convenient as its pseudopotential lies roughly midway between the cathodic and anodic stability limits which we find for the BEG-1 based electrolytes.

Cell construction.—The composite cathode was coated onto a Ni foil by using 25 w/o of the polymer gel electrolyte described above as the binder of a 9:1 (by weight) mixture of LiMnO_2 and acetylene black. The mass density of the active material of cathode film was estimated to be 8 mg/cm^2 . The carbonaceous anode film with carbon content of 8 mg/cm^2 was obtained by coating the slurry onto Ni foil.

Lithium cells Li/BEG-1 electrolyte/ LiMnO_2 and lithium-ion cells C/BEG-1 electrolyte/ LiMnO_2 were assembled in an argon dry box using 1.0 m LiClO_4 solution in 1:2 BEG-1:EC mixture with a silica fiber separator.

Using a homebuilt tester, batteries were cycled at a constant charge/discharge current density of 0.2 mA/cm^2 with upper limit 4.3 V and lower limit 2.0 V (hereafter called 4.3/2.0 V) for Li/ LiMnO_2 battery and 4.4/2.5 V for C/ LiMnO_2 battery at 28°C. Charge/discharge (coulombic) efficiency of the batteries is taken as the time ratio of discharge to charge process.

Results and Discussion

Lithium metal-electrolyte solvent stability/stabilization tests.—Referring to the tests described in the previous section, it was found that the lithium strips in pure PC formed many black patches, presumed to be PC decomposition products, while the samples in BEG-containing mixtures remained shiny; thus, 10 w/o BEG-1 provides good protection. The following two-step test suggests this is due to protective film formation: a fresh lithium foil piece was first inserted into pure BEG-1 at room temperature for 24 h and then transferred to pure PC, followed by heating at 95°C for 24 h. The lithium foil retained its original shiny appearance and maintained it at room temperature thereafter. The manner in which the BEG-1 protects lithium is not clear yet (an amorphous LiBO_2 single ion conducting film is suspected), and further work is necessary to clarify the mechanism and to optimize the condition for BEG-1 protection of lithium in organic liquid electrolytes.

Protection of lithium in various inorganic electrolyte solvents by BEG-1 was tested. The fresh lithium foils were first inserted into POCl_3 , SOCl_2 , SO_2Cl_2 , and $\text{CH}_3\text{SO}_3\text{Cl}$ vial samples and heated at 95°C (the SOCl_2 and SO_2Cl_2 have boiled) for 24 h. As a result, the surfaces of all four lithium foils were covered with a deeply colored film, which

has been identified as a polycrystalline LiCl passivation layer.^{4,5} (This is why lithium batteries using inorganic liquid electrolytes, such as Li/SOCl₂ and Li/SO₂Cl₂ batteries, always suffer from the "voltage delay" after long-term storage.) In contrast, when 20 to 30 w/o of BEG-1 was first added to the inorganic solvents, the deeply colored film did not form, presumably because it was pre-empted by a BEG-based film.

The utility of BEG-1 for improving the reversibility of initial intercalation of lithium ions into carbon was also given preliminary study. It has been well known that common organic solvents, such as PC and EC, are easily coinserted into carbon layers and may then be decomposed on the interface or surface of carbon layers when lithium is intercalated.²²⁻²⁴ As a result, much irreversibility of lithium ion intercalation into carbon layers happens during the initial cycle, since the decomposition of intercalated solvent consumes a large number of electrons. Using the cell described previously, we found that an Li/C battery using 1.0 *m* LiClO₄ solution in 1:1 DME:EC mixture has only 41% of charge/discharge efficiency in the first cycle. However, the same battery using 1.0 *m* LiClO₄ solution in 1:2 BEG-1:EC mixture demonstrates 54% efficiency in the first cycle. The reason for this difference might be that BEG-1 is preferably absorbed into the fresh carbon layers and hence protects against decomposition of EC on the carbon layers. Thus, BEG-1 may not only improve the stability of lithium metal in contact with organic/inorganic electrolyte solvents for lithium batteries but may also improve the stability of organic solvents in the presence of fresh carbon structure for the lithium-ion batteries using carbonaceous anode.

Ionic conductivity of BEG-1 electrolytes.—The state of the solutions obtained from dissolution of lithium salts varies between viscous liquid and glass-like, depending on the concentration and the kind of salt dissolved. Most salts, once dissolved, will not recrystallize from the BEG-1 solutions, either upon cooling or, within limits, on increasing the salt concentration. In other words, BEG-1 forms very stable, supersaturated solutions, which is an advantage for practical applications where low temperatures or high concentrations are involved.

Figure 1 shows the relationship between isothermal ionic conductivity between 30 and 150°C and salt concentration for (a, top) LiClO₄ and (b, bottom) LiN(SO₂CF₃)₂ solutions in BEG-1. Both solutions have a maximum conductivity of around 10⁻⁴ S/cm at room temperature falling in the mole fraction range of 0.1 to 0.2. This phenomenon is common to most electrolyte solutions and is due to the competition between the number and the mobility of charge carriers which vary oppositely with salt concentration. The latter is usually determined by the viscosity of the solution. Compared with many organic solvent-based electrolytes, BEG-1 electrolytes have low conductivity because of a much higher viscosity.

At high concentrations, a new regime is encountered for LiClO₄ solutions (see Fig. 1a). When the molar fraction of LiClO₄ exceeds 0.5, the conductivity begins to go up, even though the viscosity is also increasing rapidly. The room temperature solutions seem almost glass-like, but the conductivity reaches 10^{-4.7} S cm⁻¹ before crystallization of LiClO₄ prevents further study. This is similar to behavior reported earlier in polypropylene oxide solutions of LiClO₄,³⁶ where it was interpreted to be due to the onset of decoupled cation motion, hence of increasing Li⁺ transport numbers. The imide solutions are of uniformly higher conductivity than the perchlorates, up to mole fraction 0.5 (e.g., 10^{-5.0} vs. 10^{-5.3} at 50°C at this composition); but consistent with the poor conductivity of the pure imide (in which the Li⁺ apparently is coupled to the nitrogen), the conductivity does not show any tendency to rise again at high concentrations.

Figure 2 compares the ionic conductivities of relevant 1.0 *m* BEG-1 lithium salt electrolytes. Conductivities of LiClO₄, LiCF₃SO₃, and LiN(SO₂CF₃)₂ solutions in BEG-1

were evaluated to be 6.5 × 10⁻⁵, 1.2 × 10⁻⁴, and 1.6 × 10⁻⁴ S/cm at 25°C, and 5.9 × 10⁻⁸, 1.4 × 10⁻⁶, 6.0 × 10⁻⁷ S/cm at -20°C. The curved Arrhenius plots are characteristic of noncrystallizing electrolytes, and the low-temperature behavior suggests that the triflate solution has the lowest glass transition temperature. This must be due to the same tendency to greater ion pairing in triflate solutions seen in earlier comparisons of triflate and perchlorate solutions.³⁷ This crossover in conductivity at ~72°C may be compared with that at 105°C in the case of solutions in polypropylene oxide (Mw 4000), where it was used to estimate the relative ion pair content. We note that the 1 *m* BEG-1 solutions at 72°C are nearly one order of magnitude higher conductors than the best conducting LiClO₄ or LiCF₃SO₃ solution in PPO at that temperature,³⁷ despite comparable viscosities. This must be due to higher cation contributions to the conductivity due to BEG-1 anion interactions. The even higher conductivities of the imide solutions particularly at high temperatures, are to

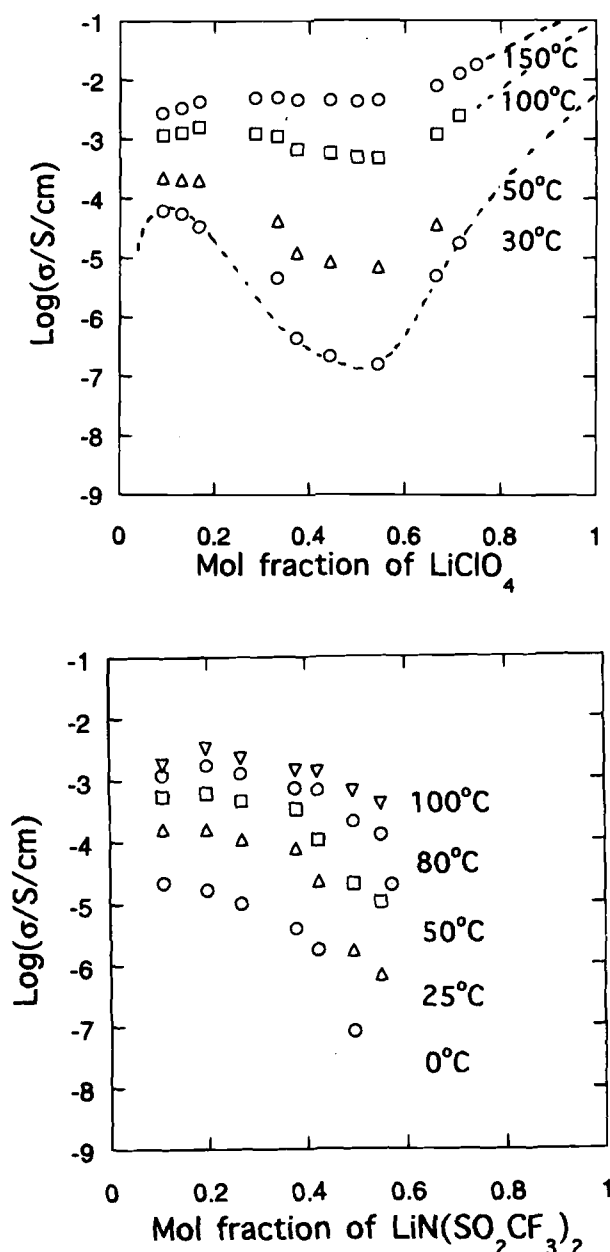


Fig. 1. Concentration dependence of isothermal conductivity for the lithium salt solutions in pure BEG-1: (a, top) LiClO₄ and (b, bottom) LiN(SO₂CF₃)₂.

be interpreted likewise. This unusual result might be ascribed to the electron-attracting property of boron atoms in the BEG-1. The center of negative charge in the $\text{N}(\text{SO}_2\text{CF}_3)_2^-$ anion is the nitrogen, the negative charge being dispersed by two electron-attracting $-\text{SO}_2\text{CF}_3$ groups. The nitrogen remains the basic site which should be preferentially coordinated by the deficit-electron BEG-1, further freeing the Li^+ to establish a higher conductivity.

In order to enhance conductivity of the BEG-1 electrolytes, it is necessary to decrease viscosity by adding other solvents. In Fig. 3 we show the conductivity of 2.5 *m* LiClO_4 dissolved in mixtures of BEG-1 and EC with different mass ratios of BEG-1 to EC. The conductivity of BEG-1:EC electrolytes rises rapidly with EC content up to ~66 w/o [83 mole percent (m/o)] but beyond this the change is smaller. In fact, electrolytes with 1:2 and 1:3 weight ratios of BEG-1:EC have almost equal conductivity (1.1×10^{-3} S/cm at 25°C). Further increase in EC content is accompanied by a decrease in electrochemical stability (discussed later). On this basis, the 1:2 weight ratio of BEG-1:EC is considered to be the optimum composition for the BEG-1:EC electrolyte solutions.

It should be noted that for 2.5 *m* LiClO_4 solution in pure EC, a freezing of solvent pure EC at around 10°C is inevitable and leads to a dramatic drop in conductivity because of the relatively higher melting point (37 to 39°C) of EC. However, there is no sign of crystallization for any of the BEG:EC solutions in Fig. 3; all curves are of the continuous curvature type even at a 1:8 weight ratio of BEG-1:EC (95 m/o EC). Thus even 5 m/o BEG-1 has a strong inhibiting effect on the crystallization of EC during cooling, implying strong EC-BEG interactions (evidenced further below).

A similar suppression of EC crystallization can be obtained by replacing some EC by PC. For instance, Fig. 4 compares the conductivity of 1.0 *m* LiClO_4 solution in pure EC and pure BEG-1, with that in 1:2 BEG-1:EC, and one in which 50% of the EC has been replaced by PC, i.e., 1:1 BEG-1:EC:PC. The electrolyte using pure BEG-1 solvent shows much lower conductivity, while that using pure EC solvent freezes at around 15°C. Unexpectedly, freezing of EC still is visible in the electrolyte using 1:2 BEG-1:EC mixture, differing from the result of Fig. 3 where a higher concentration (2.5 *m*) is involved. The freezing of EC in the 1.0 *m* LiClO_4 solution is completely avoided in the ternary

solvent down to -50°C. Thus, a combination of BEG-1 and PC is effective in eliminating the freezing of EC in EC-containing electrolytes but unfortunately also has a deleterious effect on conductivity.

The effect of both kinds and concentrations of salts on conductivity of BEG-1:EC electrolytes is shown in Fig. 5. The freezing of EC occurs in all three 1.0 *m* salt solutions but disappears in 2.5 *m* solutions. The conductivities of 1.0 *m* electrolytes in a 1:2 BEG-1:EC mixture at 25°C are 6.2×10^{-3} , 3.2×10^{-3} , and 1.9×10^{-4} S/cm for LiClO_4 , $\text{LiN}(\text{SO}_2\text{CF}_3)_2$, and LiCF_3SO_3 , respectively. The order of conductivity differs in an interesting manner from that observed from pure BEG-1 electrolytes (see Fig. 2). The reason for this difference might be the action of polar EC on the solvation of lithium cations, which strengthens the dissociation of salts and then increases the ionic conductivity, probably accompanied by a decrease in the transference number of lithium ions.

For applications it is desirable to rubberize, i.e., "gel,"³⁸ the electrolyte by small additions of entangling (high Mw) polymer. Polymers suitable for this purpose include poly(ethylene oxide) (PEO), poly(vinyl acetate) (PVAc), and PMMA. It has been found that 10 to 20 w/o depending upon desired performance temperature of the gel electrolytes, of these polymers is enough to yield solids with 1:2 BEG-1:EC liquid electrolytes which retain their (5 s) shear modulus to well above ambient. Figure 6 illustrates the conductivity of BEG-1/EC liquid electrolyte and related "gel" electrolytes. Although EC freezes out of 1.0 *m* LiClO_4 1:2 BEG-1:EC solutions at around 15°C, the crystallization is completely avoided in the gel electrolytes except for that using PEO. PEO evidently does not interact strongly with the solvents, and the solution conductivity is accordingly least affected by its presence. The higher conductivity may also reflect the low chain stiffness character of the PEO, hence smaller increase in microviscosity. Conductivities for 1.0 *m* LiClO_4 liquid electrolyte and 10 w/o of PEO, PVAc, or PMMA gelled electrolytes are 6.2×10^{-3} , 1.7×10^{-3} , 1.3×10^{-3} , and 5.0×10^{-4} S/cm at 25°C, and 3.2×10^{-5} , 4.1×10^{-6} , 6.0×10^{-5} , and 1.7×10^{-5} S/cm at -20°C. Interestingly enough, replacement of EC by PC in the PMMA gel electrolytes has a beneficial effect on conductivity, cf. Fig. 4. In the gel case, this substitution increases the conductivity at 25°C to 1.0×10^{-3} from 5.0×10^{-4} S/cm.

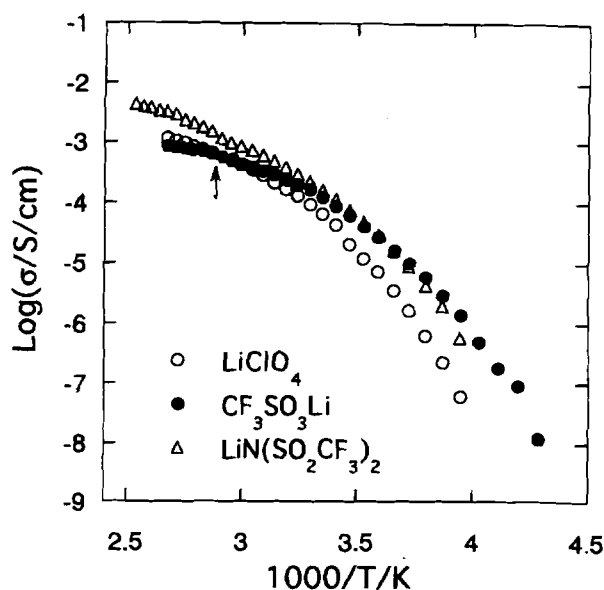


Fig. 2. Comparison of conductivity for 1.0 *m* lithium salt solutions in BEG-1.

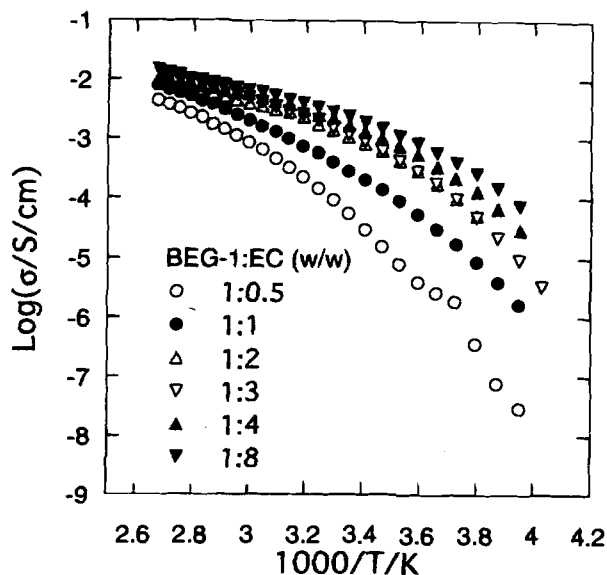


Fig. 3. Composition effect of solvents on the conductivity of 2.5 *m* LiClO_4 dissolved in the 1:*x* BEG-1:EC mixtures (by weight), where *x* = 0.5, 1, 2, 3, 4, and 8, corresponding to 0.42, 0.27, 0.15, 0.11, 0.83, and 0.43, respectively, of the mole fraction of BEG-1.

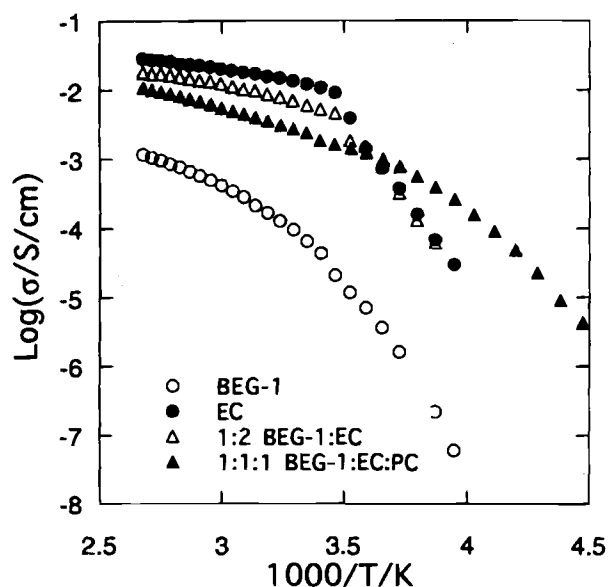


Fig. 4. Ionic conductivity of 1.0 m LiClO_4 solutions in BEG-1, EC, 1:2 (15:85) by mole) BEG-1:EC and 1:1:1 (16:45:39 by mole) BEG-1:EC:PC mixtures.

Electrochemical stability of BEG-1 electrolytes.—The electrochemical windows of BEG-1 electrolytes determined using a platinum pseudoreference electrode are shown in Fig. 7. We used Pt after determining that a 3.06 V potential difference was obtained for the cell $\text{Li}/1.0\text{ }m\text{ } \text{LiClO}_4$ in 1:1 DME:EC mixture/Pt. This value is usually taken as the potential shift for adjusting the difference between cyclic voltammograms obtained by lithium and platinum reference electrodes. Except for the potential shift, no other difference is observed between the two kinds of cyclic voltammograms. Therefore, in this work the results from the platinum pseudoreference electrode were used to analyze the electrochemical stability of the electrolytes.

Figure 7a presents cyclic voltammograms of LiClO_4 solutions in pure PC and 1:2 BEG-1:PC mixture, respec-

tively. At negative potentials, similar results are observed for each of these electrolytes (except for the conductivity-determined slope of the plot after lithium deposition commences at around -3.0 V. On the other hand, there is an important difference between the two electrolytes at positive potentials. In the case of PC electrolyte, the cathodic current is sharply increased at around 0.9 V because of the oxidation of PC. The BEG-1:PC electrolyte, on the other hand, exhibits a negligible background current until 2.2 V, implying that an electrochemical window of the BEG-1:PC electrolyte has been extended to 5.2 V from the value of 4.0 V for the pure PC electrolyte. The electron-deficient centers of the BEG molecules are evidently protecting the PC against oxidation by interacting with the electron-rich (oxidation-sensitive) carbonyl group. Such interactions would replace the BEG-BEG interaction illustrated in the introduction. Clearly the cathodic limit of the pure PC electrolyte solution is controlled by the oxidation of solvent rather than of anions.

To minimize the effect of ionic conductivity of the electrolyte on the peak current of cyclic voltammograms, an EC solution containing a higher salt concentration, $3.1\text{ }m$ LiClO_4 , was made. This concentration was chosen to yield a conductivity comparable with that of $2.5\text{ }m$ LiClO_4 solution in 1:2 BEG-1:EC mixture. In Fig. 7b we display the cyclic voltammograms of these solutions, which now have similar anodic peak currents for deposition/stripping of lithium at around -3 V. As seen before, the use of BEG-1 co-solvent increases the electrochemical window of EC electrolyte by 2.3 V, where as pure EC electrolyte decomposes at 0.7 V.

Figure 7c compares the effect of electrolyte component and salt anion on the electrochemical windows. $1.0\text{ }m$ LiClO_4 solution in 1:1 DME:EC mixture has an extremely high conductivity, 1.3×10^{-2} S/cm at 25°C , hence permitting much higher anodic currents at around -3 V for our chosen scan speed. However, this electrolyte decomposes at 1.3 V, which means the electrolyte has an electrochemical window of only 4.3 V. In contrast, $2.5\text{ }m$ LiClO_4 and $\text{LiN}(\text{SO}_2\text{CF}_3)_2$ solutions in the 1:2 BEG-1:EC mixture exhibit remarkable stabilities, 5.8 V or more. Thus BEG-1 is effective in improving the electrochemical windows of nonaqueous organic liquid electrolytes, the mechanism

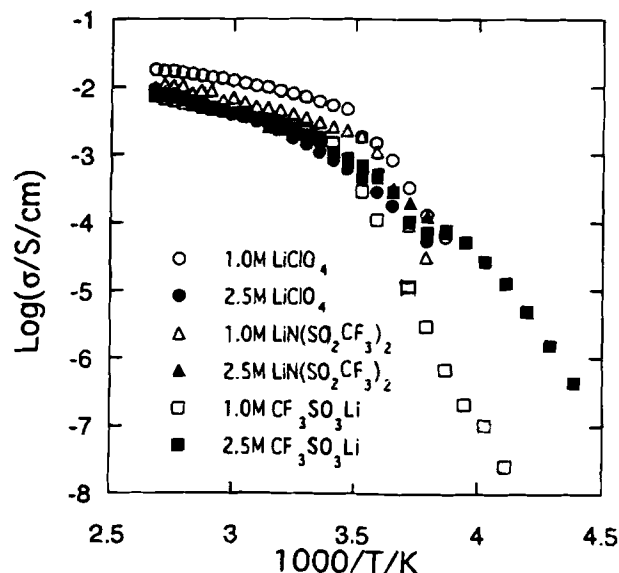


Fig. 5. Ionic conductivity of 1.0 (hollow symbol) and 2.5 m (solid symbol) LiClO_4 (circle), $\text{LiN}(\text{SO}_2\text{CF}_3)_2$ (triangle), and $\text{CF}_3\text{SO}_3\text{Li}$ (square) dissolved in 1:2 (15:85 by mole) BEG-1:EC mixture.

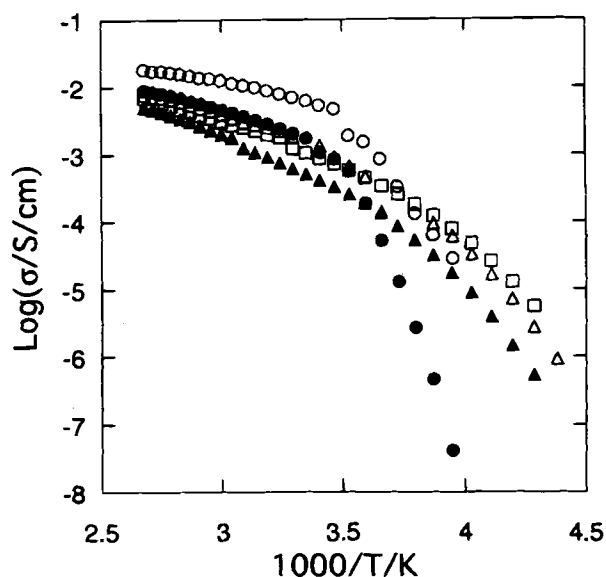


Fig. 6. Arrhenius plots of conductivity for $1.0\text{ }m$ LiClO_4 solution in 1:2 (15:85 by mole) BEG-1:EC mixture (hollow circle), the gel electrolytes formed by participating 10 w/o PEO (solid circle), PVAc (hollow triangle), and PMMA (solid triangle) as well as the gel electrolyte formed by dissolving 10 w/o PMMA into $1.0\text{ }m$ LiClO_4 solution in 1:1:1 (16:45:39 by mole) BEG-1:EC:PC (hollow square).

being uncertain at this time but presumably involving the interaction of alkene oxygen with the boron center.

Our concern in these studies was with the voltage window, but assessments of the coulombic efficiency (CE) of the lithium deposition/stripping process have also been obtained from the cyclic voltammogram data using system software. In fast scans such as we employed, very little lithium is deposited, and this will be at the mercy of any reducible impurities near the electrode surface; for instance, no oxygen purge is possible in our cell. Thus, high efficiencies are not to be expected. In the present

cases, the coulombic efficiencies ranged from 50 to 75% with no special precaution taken. This compares favorably with the case of 1.0 *m* LiClO₄ dissolved in 1:1 (by volume) DME:EC mixture for which the CE falls below 30% for the same scanning conditions.

Tests of cells using BEG-1:EC-based electrolytes.—

Results of charge/discharge tests on cathode-limited cells constructed as described earlier are shown in Fig. 8 and 9. Figure 8 is the first four cycles of a Li/1.0 *m* LiClO₄ in 1:2 BEG-1:EC mixture/LiMnO₂ battery run with a current density of 0.2 mA/cm² at 28°C. The battery shows two discharge plateaus at about 4 and 2.2 V, respectively; the total capacity of LiMnO₂ active material is estimated to be 160 mAh/g, comparable with the result by Koetschau *et al.*³⁹ Due to the change of LiMnO₂ structure caused by the charge/discharge process, the capacity of the battery dropped 40% during the initial ten cycles, over which the higher plateau gradually disappeared with very little loss of lower plateau capacity. Notwithstanding the capacity changes, the discharge/charge efficiency of the battery remained quite high. After the initial ten cycles, the capacity remained almost constant over the next 1000 cycles.

To test the stability of the composition of electrolyte solution against longtime storage, another battery with the configuration exposed to the dry-box atmosphere was assembled to measure the short-circuit current. After the battery had been kept in the dry box for a month, the short-circuit current was remeasured and found almost equal to the fresh-assembled state. This implies that the conductivity of electrolyte are stable to longtime storage, thanks to the involatility of the BEG-1:EC mixture.

Figure 9 shows selected cycles of the cell C/1.0 *m* LiClO₄ in 1:2 BEG-1:EC mixture/LiMnO₂ cycled at a current density of 0.2 mA/cm² with cutoff voltages of 4.4/2.5 V at 28°C. Owing to the stability of the electrolyte solvent containing BEG-1, the discharge/charge efficiency of the battery in the first cycle reaches 55% and exceeds 85% for the subsequent cycles. Unlike the Li/LiMnO₂ cell, the capacity plateaus in the C/LiMnO₂ battery have disappeared because of the interaction between two reversible electrochemical processes for C and LiMnO₂ during charge and discharge; however, a slow capacity loss with cycle number is still found. We believe that the reason for the capacity loss is located in the electrode materials.

Conclusion

In conclusion, the electron-deficient BEG-1 which has favorably high boiling point but low fluidity proves to be an excellent stabilizing component of electrolytes for high-voltage rechargeable lithium and lithium-ion batter-

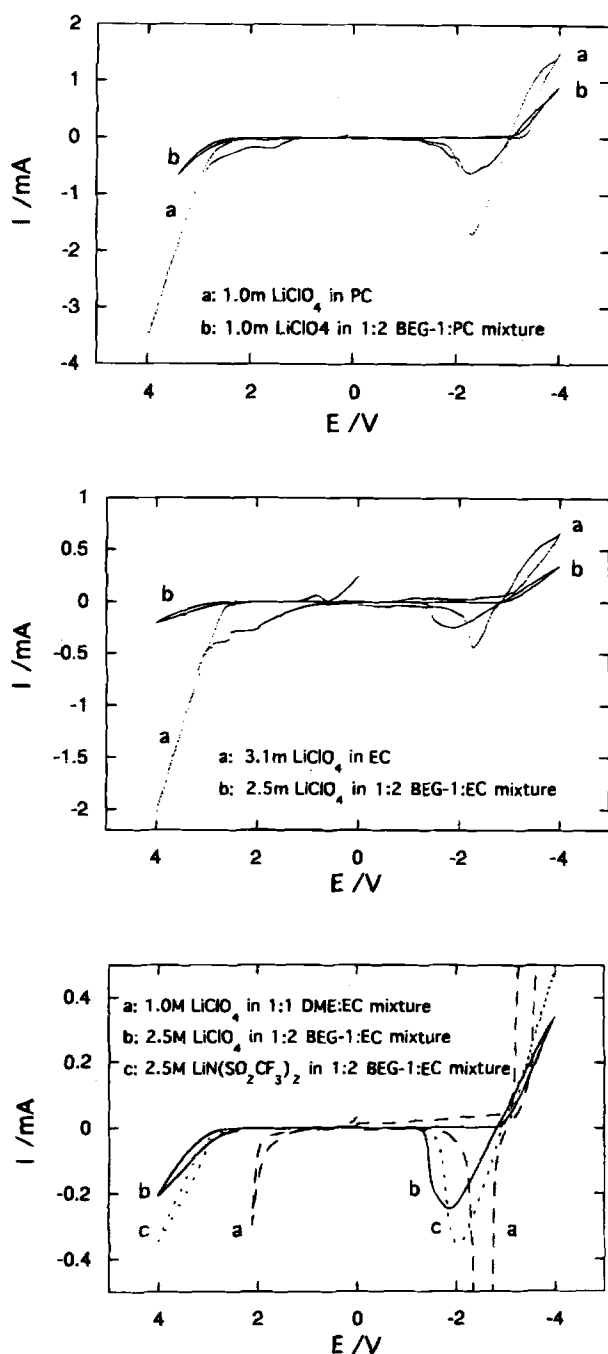


Fig. 7. Comparison of cyclic voltammograms for conventional salt-solvent solutions and BEG-1 stabilized salt-solvent solutions: (a, top) 1.0 *m* LiClO₄ in PC compared with 1:2 BEG-1:PC mixture; (b, middle) 3.1 *m* LiClO₄ in EC compared with 2.5 *m* LiClO₄ in 1:2 BEG-1:EC mixture; (c, bottom) 1.0 *m* LiClO₄ in 1:1 (by volume) DME:EC compared with 2.5 *m* LiClO₄ in 1:2 BEG-1:EC and 2.5 *m* LiN(SO₂CF₃)₂ in 1:2 BEG-1:EC mixture. (Note the almost 6 V window in the BEG-1 stabilized cases.)

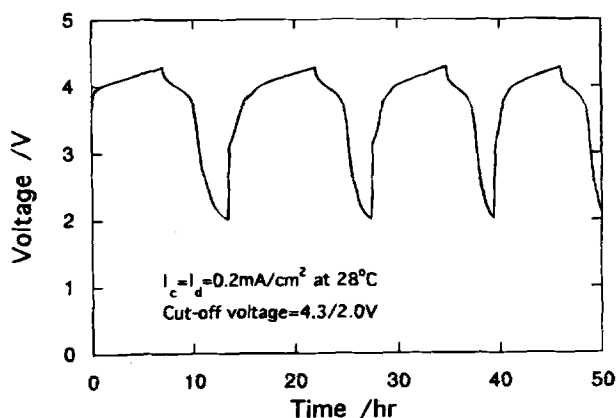


Fig. 8. First four cycles of Li/1.0 *m* LiClO₄ in 1:2 BEG-1:EC mixture/LiMnO₂ battery. $I_c = I_d = 0.2$ mA/cm², cutoff voltage 4.4/2.0 V, 28°C.

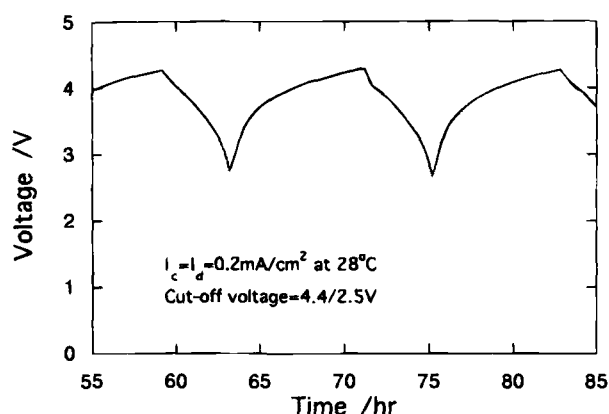


Fig. 9. Selected cycles of C/ 1.0 m LiClO_4 in 1:2 BEG-1:EC mixture/ LiMnO_2 battery. $i_c = i_d = 0.2\text{ mA/cm}^2$, cutoff voltage 4.4/2.0 V, 28°C .

ies. The ionic conductivity of the electrolytes containing BEG-1 reaches values acceptable for battery use provided organic or inorganic solvents, such as PC, EC, SOCl_2 , SO_2Cl_2 , POCl_3 , and $\text{CH}_3\text{SO}_2\text{Cl}$, are added as co-solvents. The electrolyte with a composition of 1.0 m LiClO_4 dissolved in 1:2 BEG-1:EC mixture has an ionic conductivity of $1.8 \times 10^{-3}\text{ S/cm}$ at 25°C . Using 10–20 w/o of high Mw polymer, BEG-1 electrolytes may be conveniently rubberized ("gelled") without serious loss of conductivity. More important, as an electrolyte additive, BEG-1 can effectively improve the stability of lithium to organic and inorganic electrolyte solvents and the stability of electrolyte solvents to fresh carbon surfaces. We propose BEG-1 and related solvents^{33,34} as superior electrolyte co-solvents, additives, or even pretreatment reactants for high-voltage chargeable lithium battery and lithium-ion batteries.

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